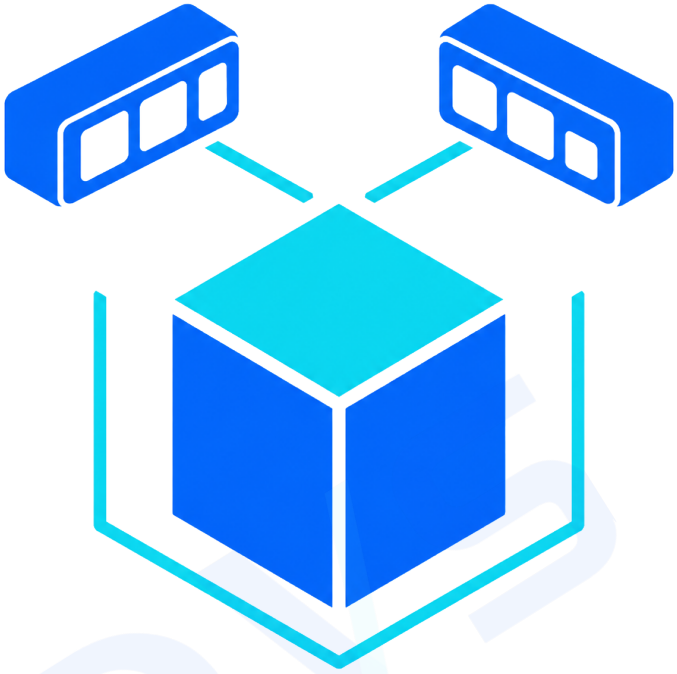
MRDVS
Mobile Robot Vision Expert



AW3 TorusMetric

Volume Measurement Solution White Paper

Measure object size and volume in 3D.

Zhejiang MRDVS Technology Co., Ltd.

V0.1 · 2026-09-09

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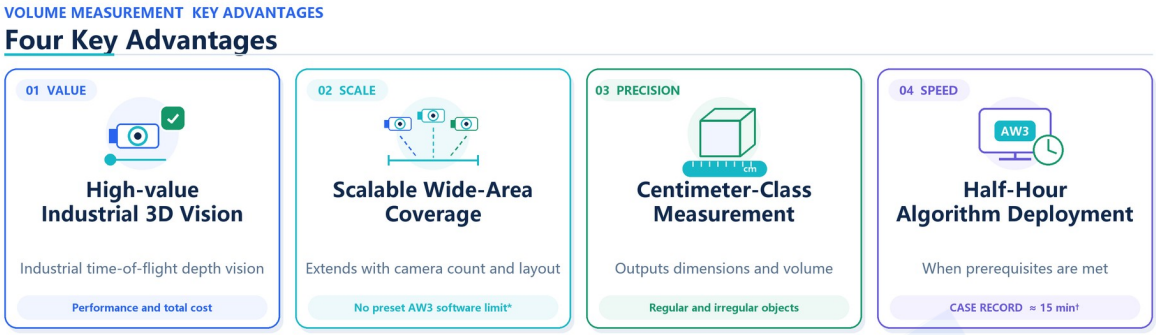
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MRDVS

1 Product Overview

Volume Measurement runs on AW3. At a fixed station, it uses 3D point clouds to output length, width, height, [Bounding Box Volume], and an optional [Real Volume] estimate for single- or multi-camera layouts.

1.1 Key Advantages

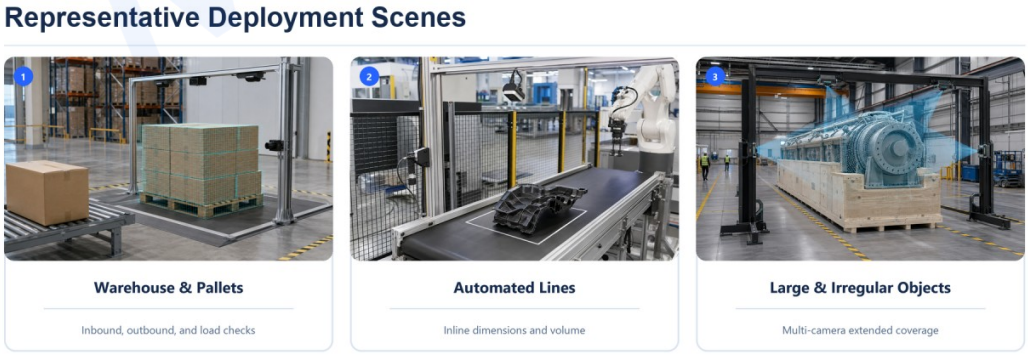


Value from 3D vision hardware to algorithm deployment

1.2 Applications



Six typical applications



Representative deployments for warehouses, automated lines, and large objects

Positioning Designed for centimeter-class dimensioning and space assessment. It is not for millimeter metrology, billing, or net material volume. Validate performance in the final layout with representative objects.

1.3 M10 RGB-D Camera

M10 is the compact industrial dToF RGB-D camera used in this solution. It provides RGB images, depth maps, IR images and 3D point clouds to AW3 TorusMetric for single- or multi-camera volume measurement.



Product Specifications	Specification
Model	M10
Sensor Type	dToF RGB-D
Laser Wavelength	940 nm
Output	Depth map / IR image / RGB image / 3D point cloud
TOF Resolution/Frame Rate	480 × 320 @ 5 fps
TOF Field of View (H×V)	90° × 60° (±3°)
RGB Resolution/Frame Rate	1920 × 1080 @ 5 fps; max. 20 fps
RGB Field of View (H×V)	90° × 60° (±3°)
Detection Range	0.5–5 m (10% reflectivity)
Measurement Accuracy	±30 mm
Average Power Consumption	≤4 W
Size / Weight	80 × 37 × 25 mm / approx. 190 g
Power Supply	12–27 V
Communication Interfaces	Ethernet
IP Rating / Eye Safety	IP54 / CLASS I
Operating Temperature	−20°C to 60°C
Storage Temperature	−40°C to 85°C
Time Synchronization	PTP
Software Support	C/C++/ROS SDK
Supported Platforms	Windows 7/8/10/11, Linux, Arm Linux/ROS
Ambient Light Immunity	100 kLux

The camera is not waterproof. Do not expose it directly to rain, snow or water spray; use it in a dry, sheltered environment. Camera ranging accuracy is distinct from solution-level dimensional error.

Source: M10 RGBD Camera Specification V1.2. FOV tolerance: ±3°.

2 Volume Measurement Principles

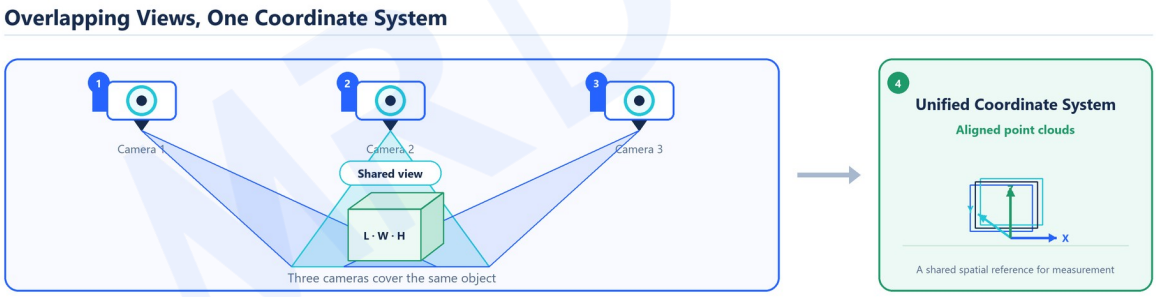
2.1 Dimension and Volume Calculation



From point-cloud capture to measurement output

Cameras capture depth point clouds. AW3 applies the camera- and ground-calibration results. Volume Measurement crops each Detection Area, removes background and noise, groups separate objects, and calculates dimensions, volume, and count. AW3 then displays or forwards the results.

2.2 Multi-Camera Stitching Principle



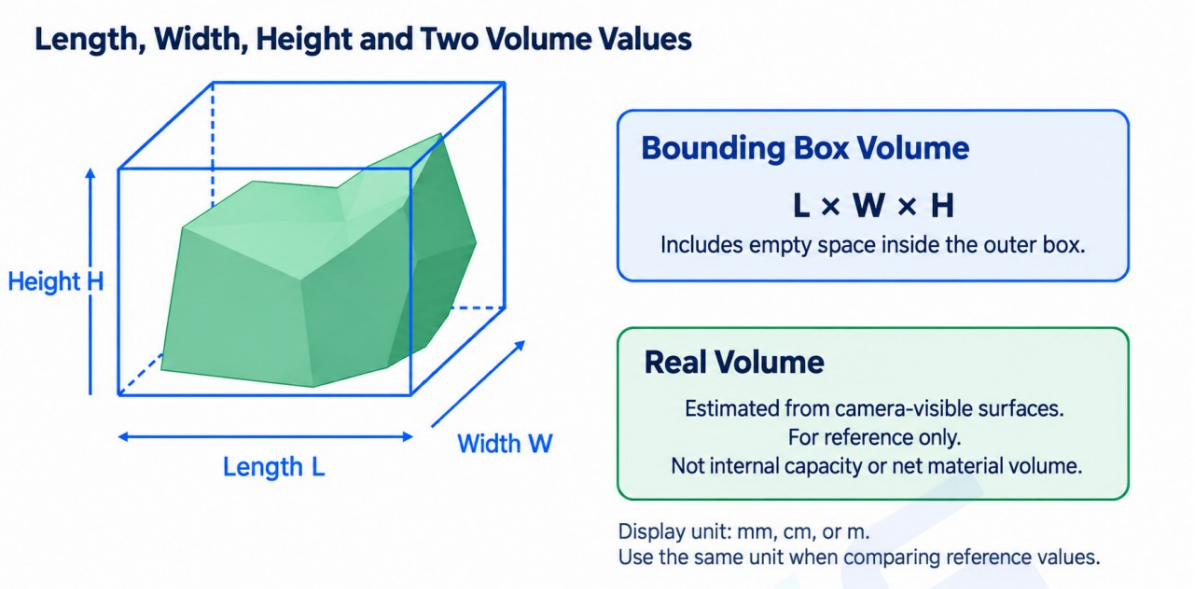
Adjacent cameras use a shared calibration board to build one coordinate system

Adjacent cameras see the same board at several positions. AW3 estimates the camera transforms and maps each depth cloud into the reference-camera coordinates. A connected camera chain extends coverage, reduces blind spots, and supports large or oversized objects.

Quality Check Use pose error, total error, and merged point-cloud overlap to review stitching. Calibration error indicates alignment quality; it is not dimensional accuracy. Skip Multi-Camera Calibration when only one camera is used.

3 Capabilities and Deployment

3.1 Measurement Outputs and Coverage

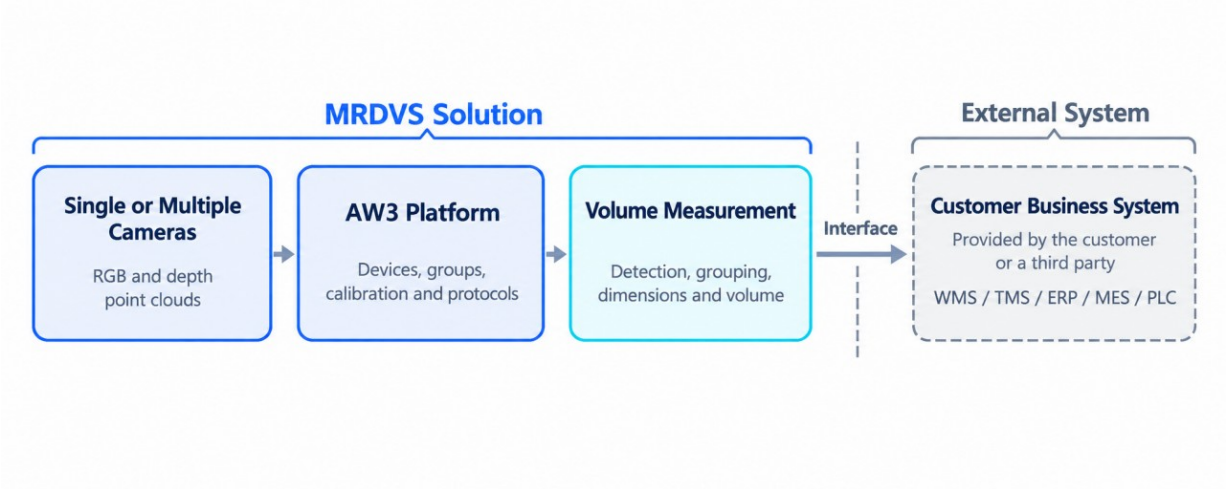


Dimensions and the two volume outputs

Capability	Definition	Business Use
Length, Width, Height	Bounding dimensions relative to the calibrated platform	Checks, packaging, and loading
Bounding Box Volume	Length × width × height, including empty space	Occupied-space assessment
Real Volume	Estimate built from surfaces visible to the cameras	Reference for irregular objects
Multi-Camera Stitching	Aligns overlapping views in one coordinate system	More coverage and fewer blind spots
Detection Areas and Multiple Objects	Supports one to five Detection Areas, with per-object or merged output	Multiple stations, zones, and grouped loads

Wide-Area Coverage AW3 does not impose a fixed software coverage size. Multi-camera layouts can be extended, but measurable coverage is project-specific. It depends on camera selection and count, layout, shared views, installation height, point-cloud quality, and acceptance criteria. [Real Volume] is estimated from visible surfaces and is for reference only.

3.2 Deployment Workflow and Key Settings



MRDVS provides the measurement solution; external business systems are provided by the customer or a third party.

- Single camera: use when one view covers the target. Skip Multi-Camera Calibration; complete Ground Calibration and Detection Area setup.
- Multiple cameras: align overlapping views into one point cloud. This is the main advantage for extending coverage to large or oversized objects.
- Mount cameras rigidly. They must not move after calibration and do not need to be level.

Key Setting	Default	Purpose
Minimum Point Count	50	Remove clusters with too few points
Detection Interval	0 ms	Interval setting; it does not mean zero latency
Minimum Neighbors	2	Filter noise by nearby point count
Neighbor Radius	3	Works with Minimum Neighbors
Real Volume Resolution	5 mm	Estimate grid size, not accuracy

About Half-Hour Algorithm Deployment When prerequisites are met, multi-camera stitching and Volume Measurement configuration can be planned at about half-hour scale. In one 11-camera case covering about 4 m × 8 m × 4 m, AW3 group setup, stitching, and Volume Measurement configuration took about 15 minutes. Timing started only after installation, power, network, valid depth data, and shared views were ready. It excludes hardware installation, cabling, site changes, and final acceptance. Actual time is project-specific.

Platform Responsibilities AW3 manages device access, camera groups, calibration, Volume Measurement settings, monitoring, and protocols. Protocol identifiers and field mappings are project configurable.

4 Performance and Scope

4.1 Standard Capabilities and Validation Results

Volume Measurement is designed for centimeter-class dimension and volume measurement. The table separates standard support, representative test results, and project acceptance. Stitching error measures point-cloud alignment, not dimensional accuracy.

Category	Metric	Value and Meaning
Standard Capability	Measurement Class	Centimeter-class; not for millimeter metrology
Standard Scope	Installation Height	Up to 4 m; assess higher installations before commitment
Standard Scope	Suggested Object Size	About 20 cm or larger; validate smaller objects first
Calibration Quality	Maximum Multi-Camera Stitching Error	≤20 mm by default; ≤25 mm only when accepted by the project; not dimensional accuracy
Representative Test	Test Setup	Indoor, 11 cameras; about 7.6 m × 3.2 m coverage; about 3.1 m height; 1.3 cm stitching calibration error. This is the test setup, not a maximum measurement range
Representative Test	Dimension Error	About 3 cm; maximum axis difference across representative samples
Representative Test	Real Volume Relative Error	About 4.8%; visible-surface estimate for reference only
Project Acceptance	Final Performance	Confirm with the final installation and representative and boundary samples; assess out-of-scope requirements before commitment

Result Basis The 3 cm and 4.8% values are results from the representative test above, not universal accuracy guarantees. Final performance depends on camera layout, point-cloud quality, calibration, object surface, occlusion, Detection Areas, and filters.

5 AW3 Integration and Data Output

AW3 brings camera connections, camera groups, multi-camera stitching, ground calibration, Volume Measurement settings, monitoring, and data output into one platform. It supports external triggers, result forwarding, and custom protocols. Create or edit protocol rules for each business system.



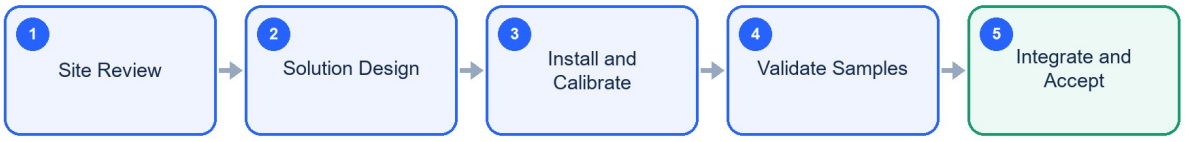
AW3 forwards results to customer or third-party business systems through configured interfaces.

Important Result	Protocol Content	Use
Length, Width, Height	length / width / height in mm	Dimensioning, packaging, and loading
Bounding Box Volume	bbox_volume_mm3 in mm^3	Occupied space for one object
Merged Volume	With Merge Area Objects on, the merged result still uses bbox_volume_mm3	One result for an area or pallet; gaps are included
Real Volume	Optional real_volume_mm3 in mm^3	Store the visible-surface estimate
Area and Count	Area ID and count	Map a station and confirm item count
Status and Time	ret_code, capture time, result time	Validate and trace results

Integration Use AW3 custom protocols, protocol identifiers, and field mapping to connect WMS, TMS, ERP, MES, PLC, or other systems. Confirm field names, triggers, and error handling for each project. This paper lists only important results, not the full JSON. Business systems are supplied by the customer or a third party and are not included in this solution.

6 Implementation and Operating Requirements

6.1 Implementation, Validation, and Acceptance



Project path from site review to acceptance

Review objects, coverage, support surface, occlusion, lighting, cycle time, and interfaces. Set camera count, shared views, and Detection Areas. After calibration, validate representative and edge samples for accuracy, repeatability, stability, time to result, and integration. Record final settings and acceptance results.

6.2 Operating Conditions and Limits

Factor	Recommended	Assess This Risk
Installation	Fixed cameras; clean lenses; no more than 4 m	Vibration, movement, or above 4 m
Support Surface	Stable and ground calibrated	Tilt, motion, or height change
Object	About 20 cm or larger; still; complete point cloud	Smaller, thin, occluded, touching, or moving
Surface and Shape	Continuous depth; mostly closed top	Black, reflective, smooth plastic, open, or inverted
Environment	Dry, sheltered indoor environment	Not waterproof: no direct rain, snow or water spray. Strong light, dust and occlusion affect measurement.

Assessment Required Outside these limits, assess the site and representative samples before commitment. Agree the acceptance criteria first.

Operations Note If streaming fails after LanxincameraViewer use, select [Restart current center] in [Device Connection / Base Camera]. Recalibrate after camera or support-surface changes.

7 Frequently Asked Questions

Question	Short Answer
What if only one camera is used?	Use the single-camera path. Skip Multi-Camera Calibration; complete Ground Calibration, Detection Areas, and sample checks.
Can it measure large or oversized objects?	Multi-camera stitching extends coverage. The measurable range is project-specific and must be validated.
Is [Real Volume] the true net volume?	No. It is an estimate from visible surfaces. It is not internal capacity or net material volume.
How does it connect to an existing system?	Use AW3 triggers, result forwarding, and custom protocols to map key fields.

Revision History and Technical Support

Version	Date	Revision
V0.1	2026-09-09	Updated 3D illustration and rain/snow restrictions; added M10 specifications, revised dimensional error to 3 cm and clarified external-system scope.

Technical Support

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